

# Quantum Communication Networks

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# ***Agenda***

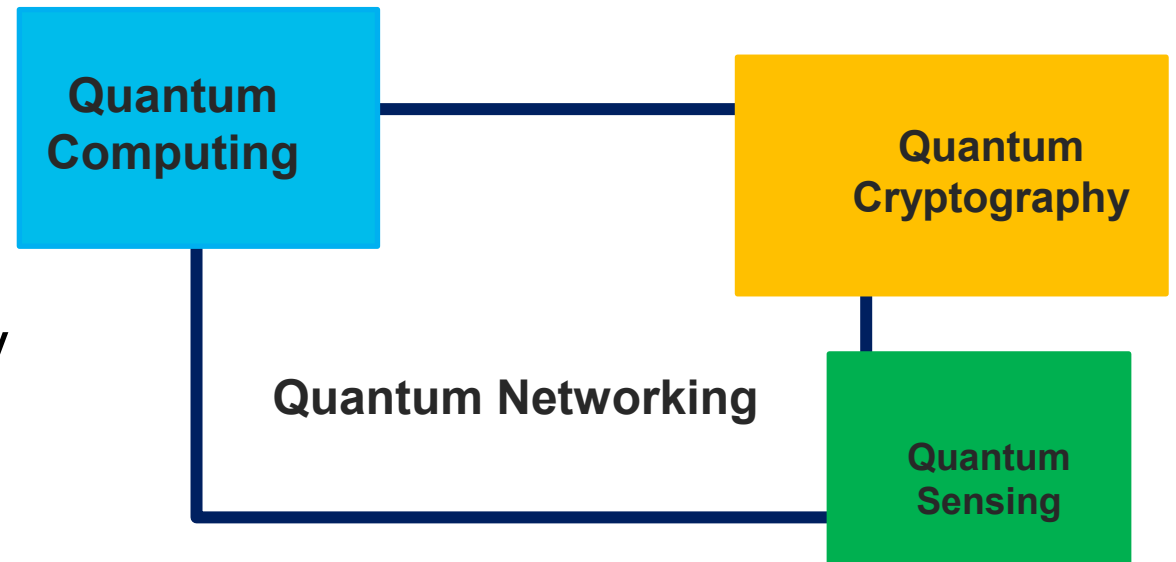
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- Introduction and Overview
- What is the Quantum Internet?
- Layered Architecture
- Current Scenario and Challenges
- Quantum Network Protocols

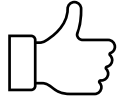
# Quantum Communication Networks

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- Quantum Communication Networks:
  - Network connecting quantum computers/nodes to transfer quantum information
- Main Motivation
  - Enable applications that are out of reach for the classical Internet
    - Secure communication
    - Powerful distributed computing
    - Advanced sensing and metrology



# Quantum market is growing



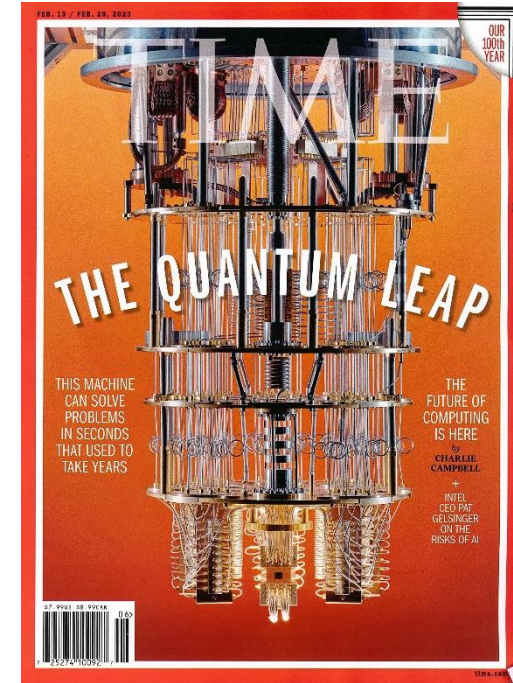
**Quantum computing and Quantum Internet projects need industry-scale investment and operation.**



Unreal promises for quantum computing applications

A flood of venture capital money

Disproportionate allocation of funds among quantum technologies.



433 qubit processor ('Osprey') by IBM

Need millions of qubits to break RSA encryption in a day or two

Courtesy: Dr. Alireza Shabani

# ***Why quantum networking and quantum internet?***

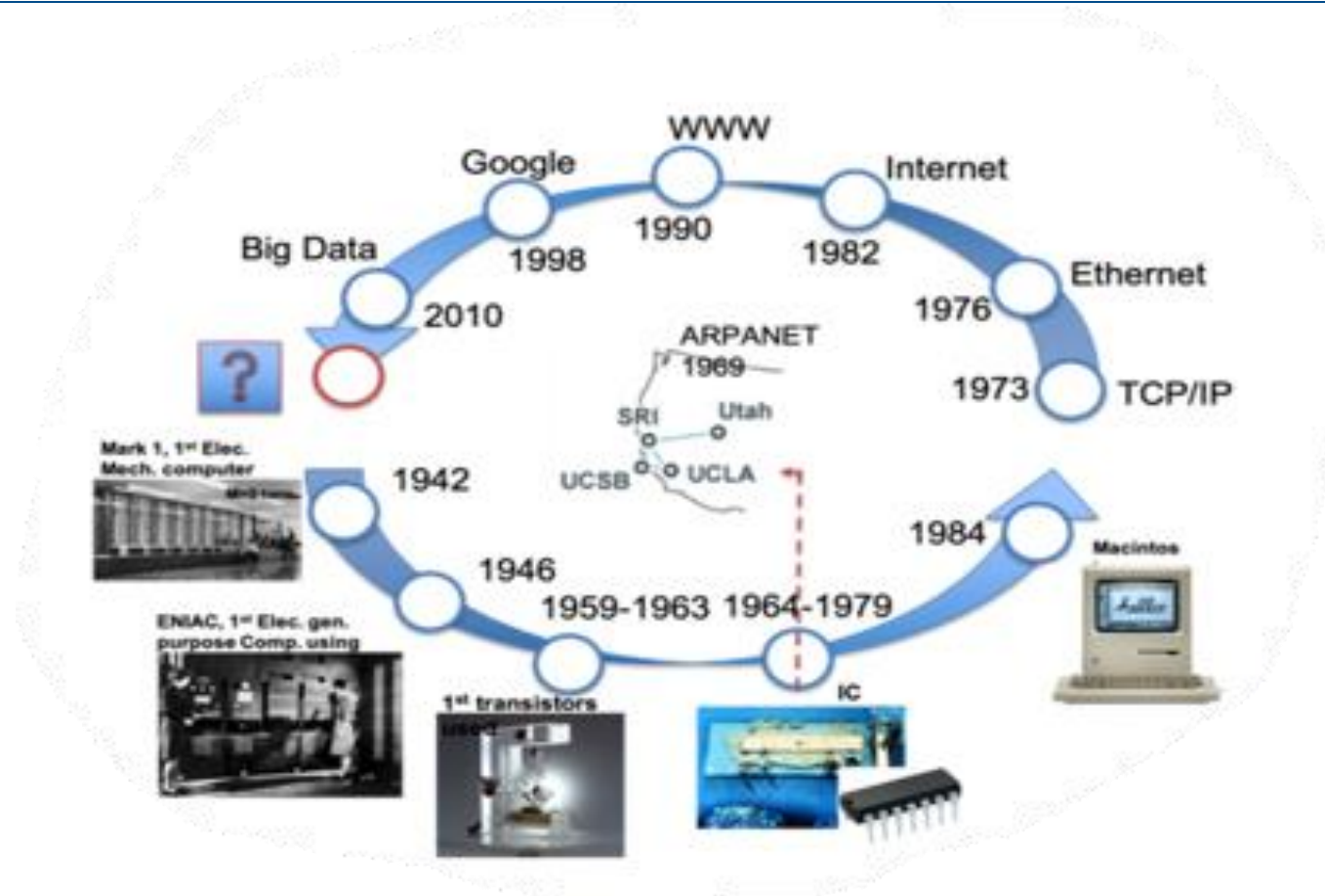
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**Internet:** distributes (qu)bits between different locations for various applications

- Today's Internet challenges
  - Best-effort services (TCP/IP)
  - Scalability
  - Data security
  - Energy per bit
  - Information rate per photon
  - Slow data processing
- Quantum Internet
  - Features:
    - Superposition, Entanglement
  - Security
  - Energy-efficient
  - Exponential speedup in processing

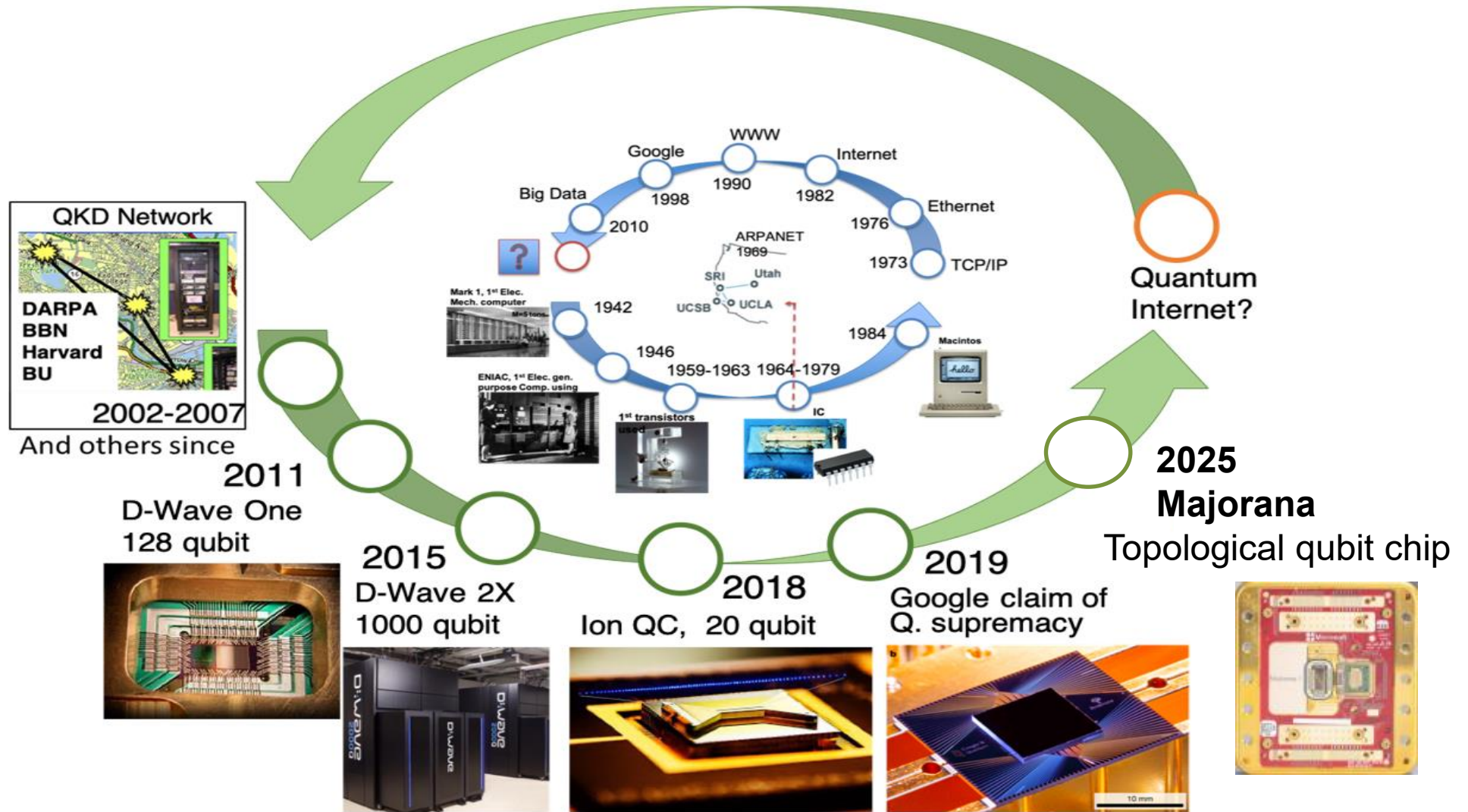
**The goal is to build quantum LANs,  
MANs, and WANs  
towards the Quantum Internet.**

# Network evolution: from technology to applications





# Network evolution: from technology to applications



Courtesy: Prof. Prem Kumar, Northwestern University, USA

# Quantum mechanics: early days to demonstration

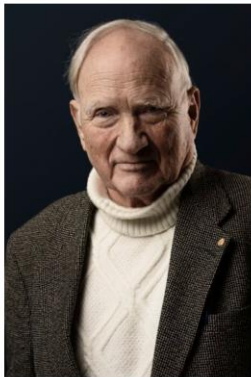
*The most incomprehensible thing about the world is that it is comprehensible.* – Albert Einstein

If you think you understand quantum mechanics, you don't understand quantum mechanics.— Richard Feynman

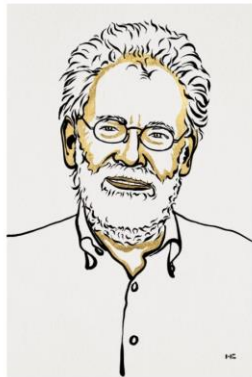
## The Nobel Prize in Physics 2022



© Nobel Prize Outreach. Photo: Stefan Bladh  
Alain Aspect



© Nobel Prize Outreach. Photo: Stefan Bladh  
John F. Clauser



Ill. Niklas Elmehed © Nobel Prize Outreach  
Anton Zeilinger

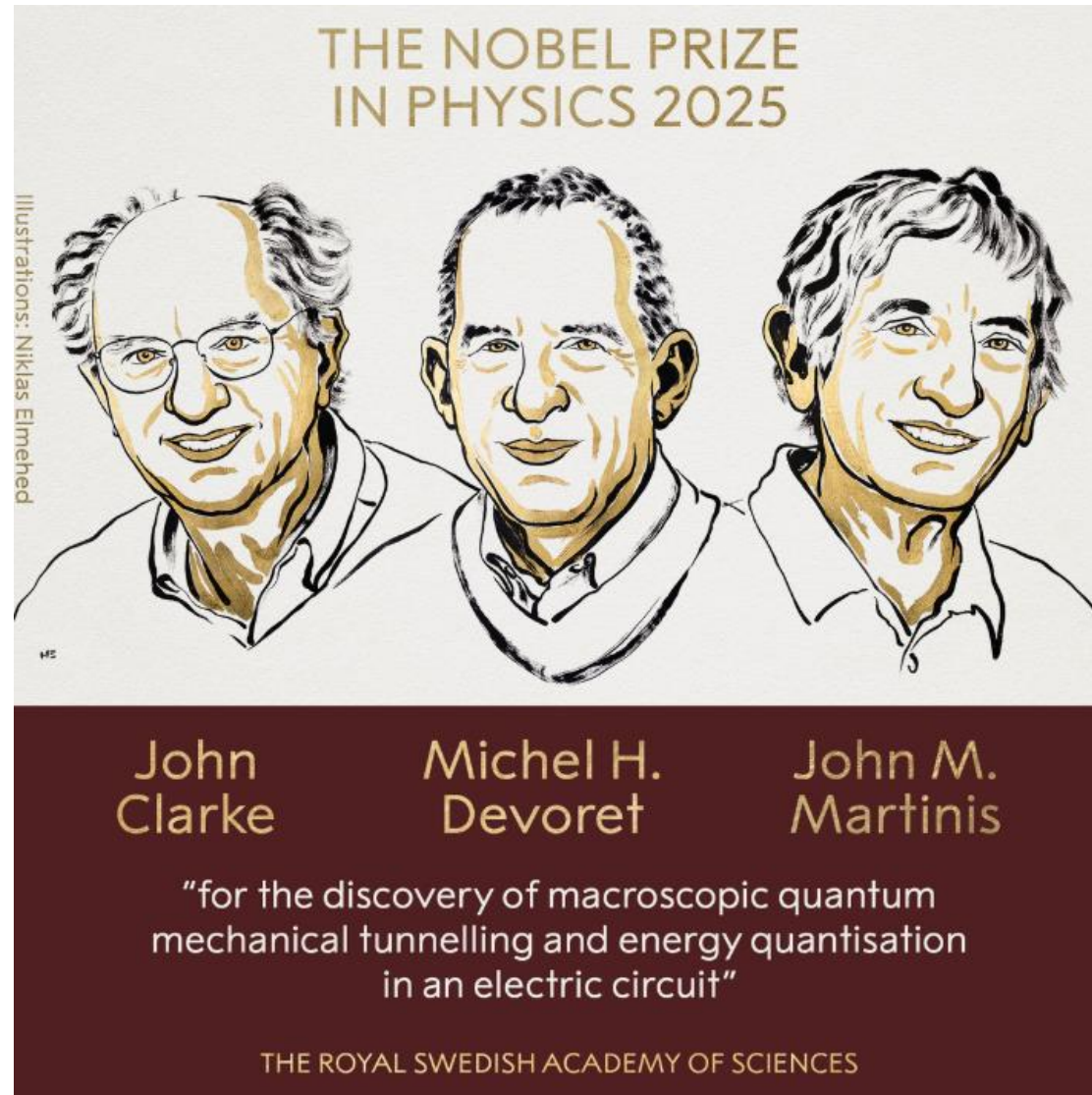
A. Aspect: for experiments with entangled photons

J. F. Clauser: for establishing the violation of Bell inequalities

A. Zeilinger: for pioneering quantum information science



# Quantum mechanics: early days to demonstration

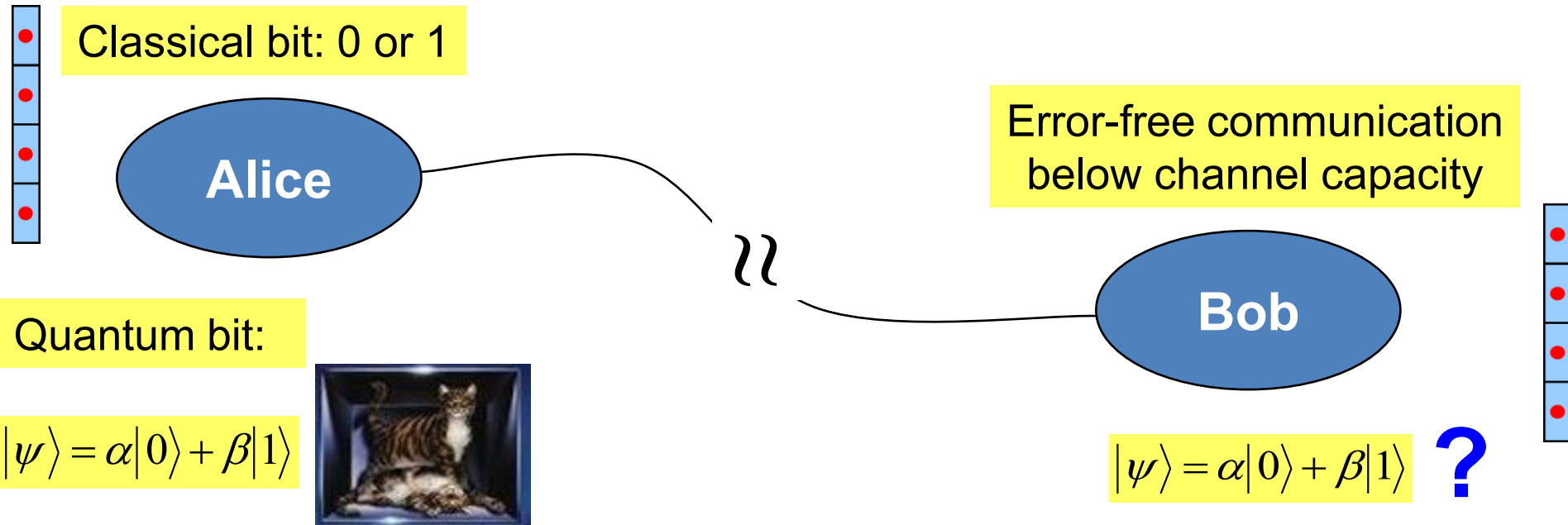


# Classical vs. quantum communication

- In a classical system, a *bit* is represented as:
  - On state = 1, Off state = 0
  - System must be in state 1 or state 0
- In a quantum system, one registers a *qubit* as:
  - Atom example: ground state =  $|0\rangle$  and excited state =  $|1\rangle$
  - Photon example:  $\uparrow$ -polarization =  $|0\rangle$  and  $\bullet$ -polarization =  $|1\rangle$
$$|H\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |V\rangle = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$
  - System can be in *superposition* state  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$  with  $|\alpha|^2 + |\beta|^2 = 1$



# Classical vs. quantum communication



## Conflict with Quantum Mechanics

- No-cloning theorem
  - It is impossible to duplicate an unknown quantum state
- Heisenberg uncertainty principle
  - It is impossible to know a quantum state

# Quantum features: superposition & entanglement

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## Superposition of qubits

$$\begin{array}{l} 0100110101 \\ +p_0 0000000000 \\ +p_1 0000000001 \\ +p_2 0000000010 \\ \dots \\ +p_{2^N} 1111111111 \end{array}$$

Particles can exist in multiple states at the same time, known as superposition

**Superposition leads to uncertainty**

# Entanglement as a unique feature of quantum communication

- Two classical systems can be *statistically dependent*

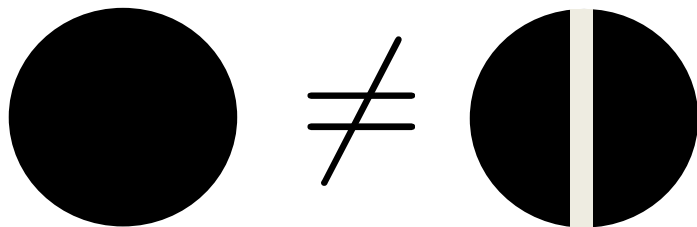
- $\Pr(\text{bit}_1, \text{bit}_2) \neq \Pr(\text{bit}_1) \Pr(\text{bit}_2)$

- Two quantum systems can be *entangled*

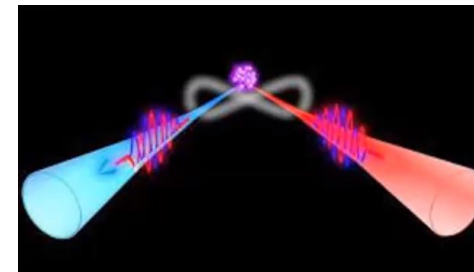
- System state: 
$$|\psi\rangle_{12} = \frac{(|0\rangle_1|1\rangle_2 - |1\rangle_1|0\rangle_2)}{\sqrt{2}}$$



*Entanglement (strange correlations)*



We know the state of the system as a whole,  
not the individual pieces



$|\text{atom cold}\rangle|\text{left photon}\rangle +$   
 $|\text{atom hot}\rangle|\text{right photon}\rangle$



# Quantum networks

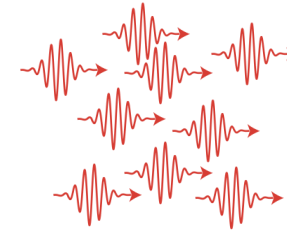
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Transferring and distributing quantum information (qubits)

Individual photons as the  
physical carriers of  
quantum information



VS



Quantum

Classical

# Quantum networks

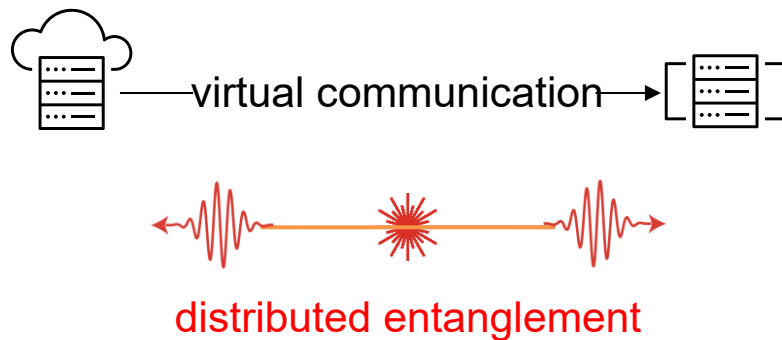
Transferring and distributing quantum information (qubits)

Individual photons as the  
physical carriers of  
quantum information



**No Cloning Theorem** – enables Security

**Teleportation**



**No Signal Amplification**

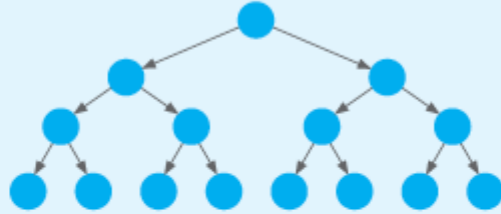


# Quantum network topologies



**Linear chain**

Each quantum node is coupled to the two nearest neighboring nodes.



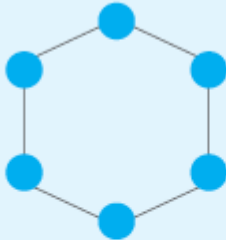
**Tree tensor**

Each node has its own hierarchical tensor with branched connections.



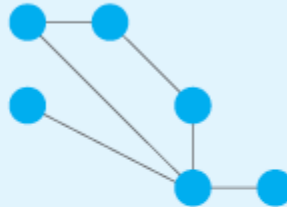
**Star**

A single node is coupled to all other nodes, but those nodes aren't coupled to each other.



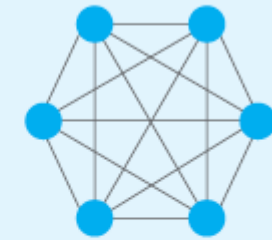
**Ring**

Each node is coupled to the preceding and successive nodes, creating a circular arrangement.



**Mesh**

A single node is coupled to several other nodes, but the connected nodes don't have to be neighbors.



**Fully connected**

Each node is directly coupled to the other nodes.

<https://www.techtarget.com/searchnetworking/tip/Types-of-quantum-network-topologies>

# Internet topologies

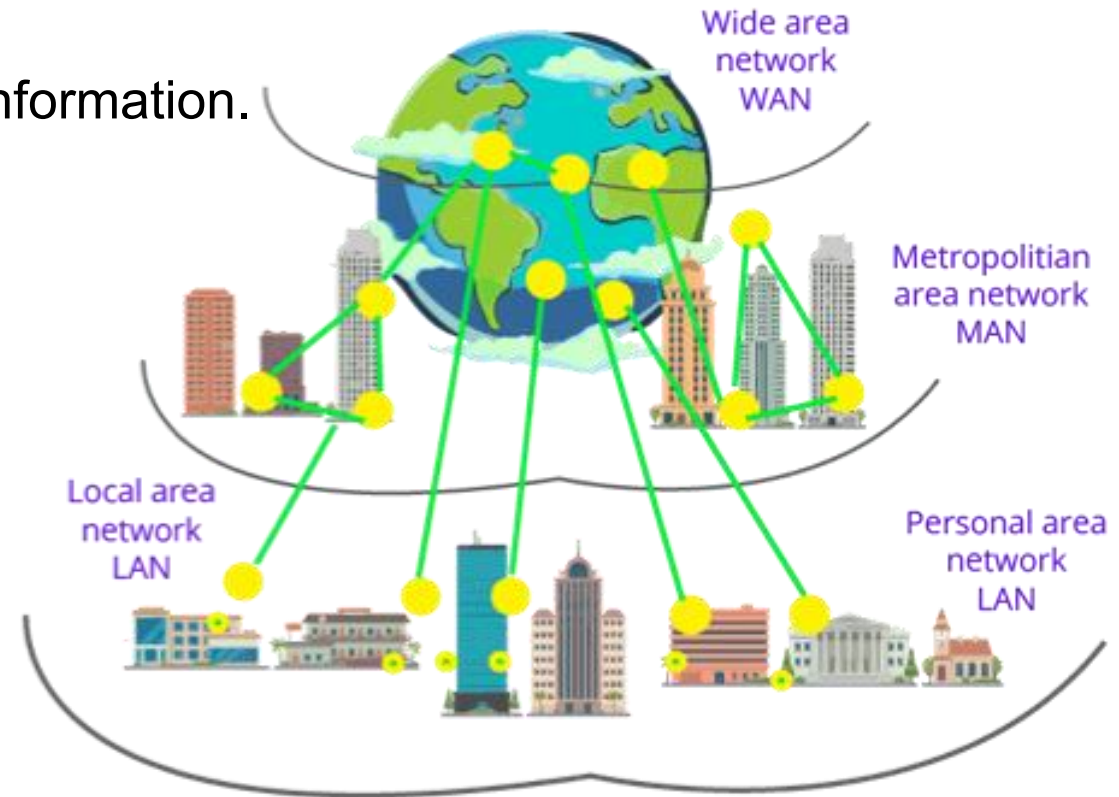
## What is the Internet?

A global network of networks to transfer bits of information.

## What is the Quantum Internet?

A global network of networks to transfer quantum bits of information,

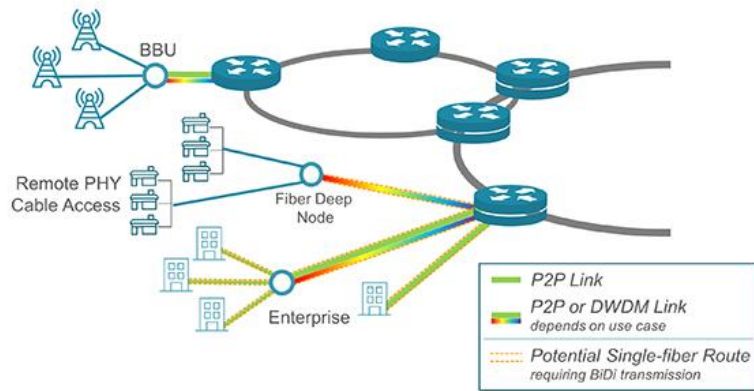
that uses the same optical medium as the (classical) internet.



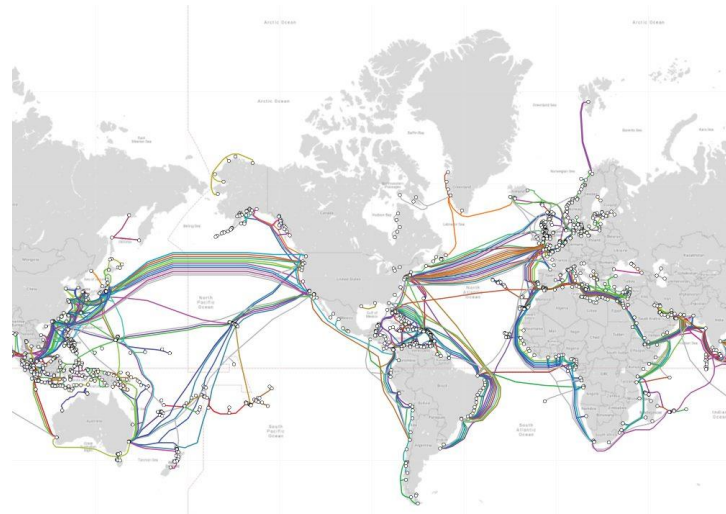
source: vendantu.com

# Classical optical networks

## Metropolitan Area



## Submarine Fibers



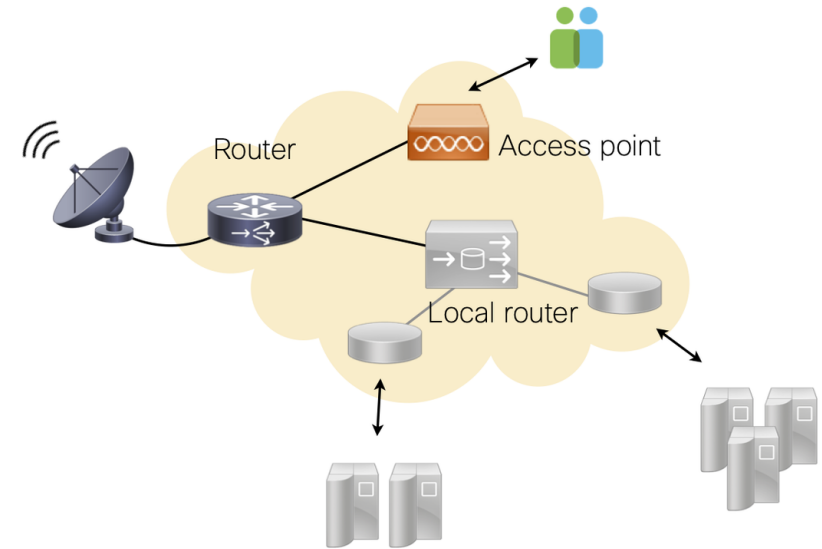
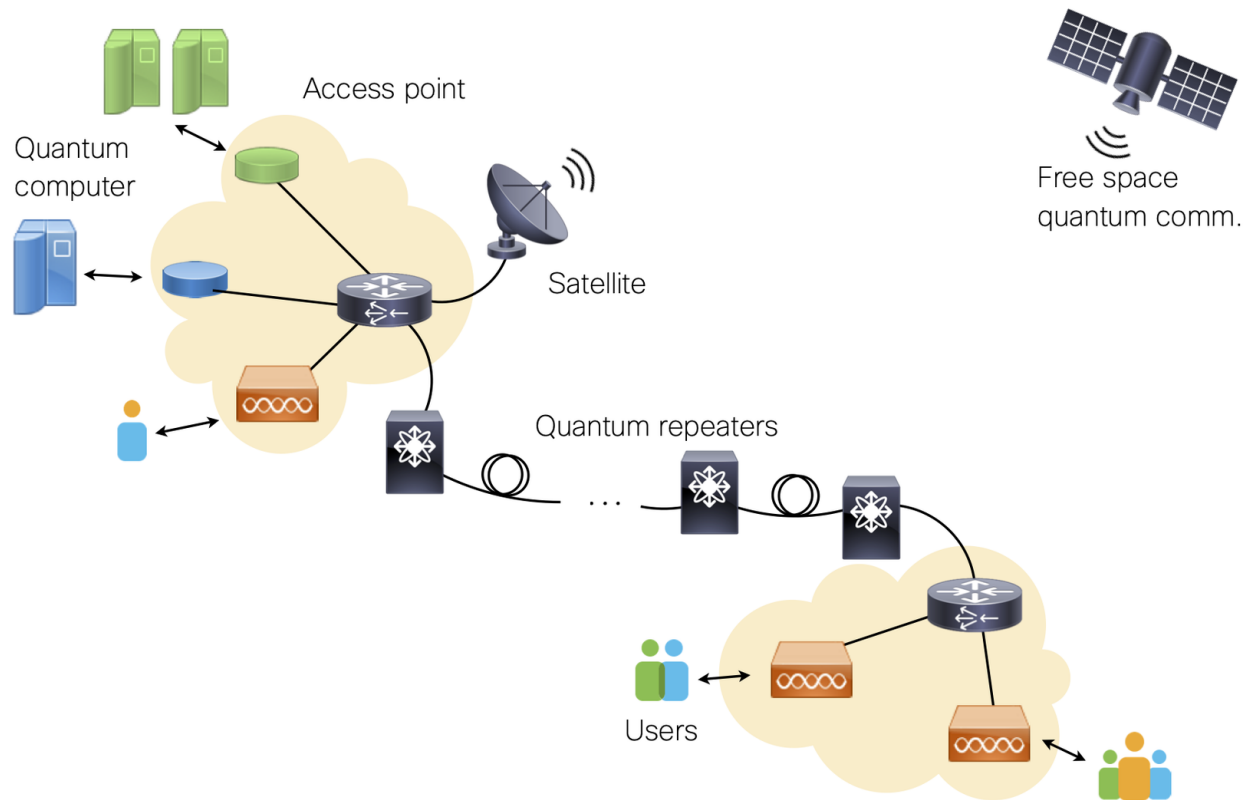
## Satellite Laser Communication



Source: TeleGeography,  
Copyright © Prisma, Inc.

Strong light signals with tens of thousands of photons

# Quantum internet: components





# Quantum Internet Testbed

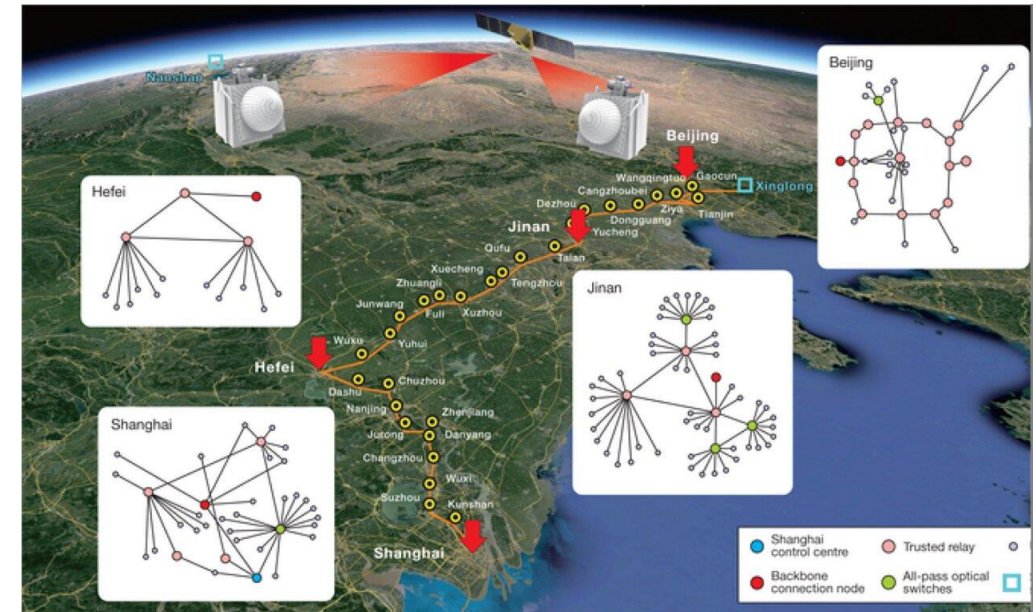
## Chicago Area



200 km network

Courtesy of Chicago Quantum Exchange

## China



4600 km network

Nature, 589, 214, 2021.

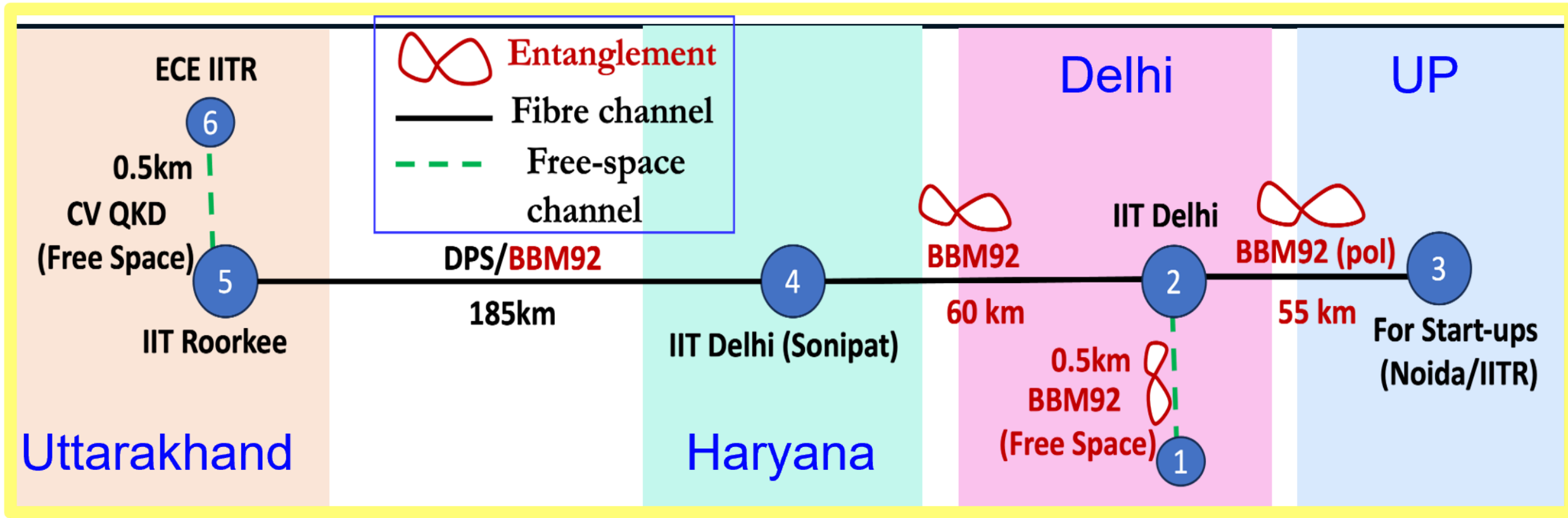
# ***Quantum communication demonstration in India***

- Space Applications Centre (SAC), Ahmedabad, between two buildings separated by 300 meters.
- BBM92 protocol implementation capable of providing secure keys for various applications.



# Proposed quantum network and protocols

## Delhi to Roorkee, 2028





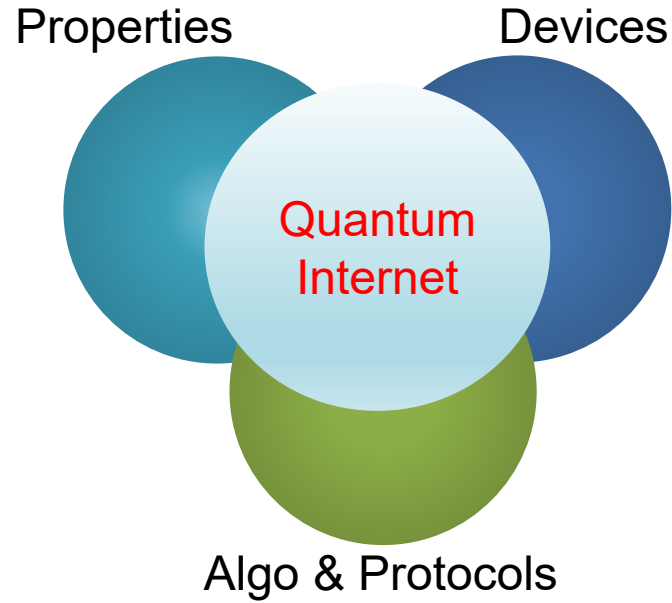
# 1000 KM Quantum Communication Demonstration



<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2250162&reg=3&lang=1>

# Quantum internet –opportunities

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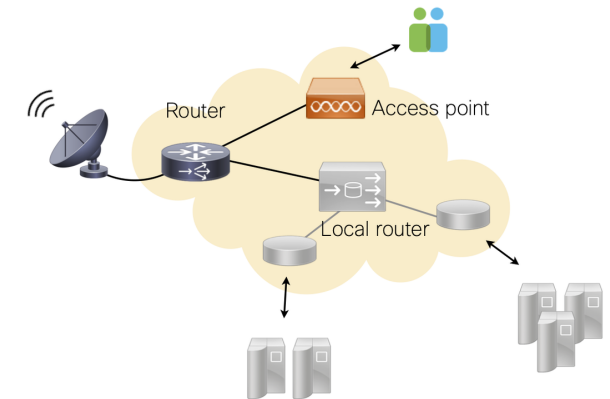
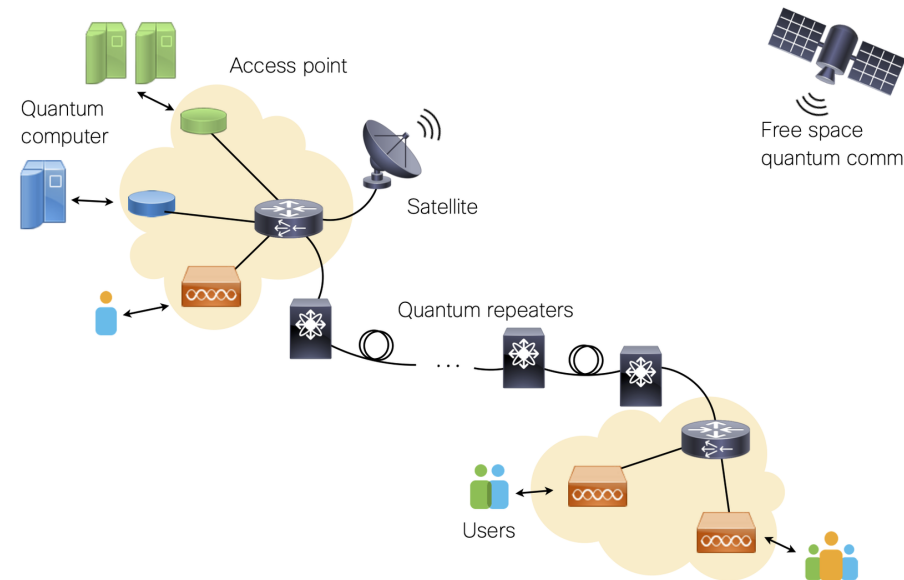


- **Opportunities to develop:**
  - Devices (Processor, memory, etc.)
  - Quantum protocols, algorithms, simulation models
  - Entanglement distribution
  - Quantum Teleportation
  - Quantum Error Correction

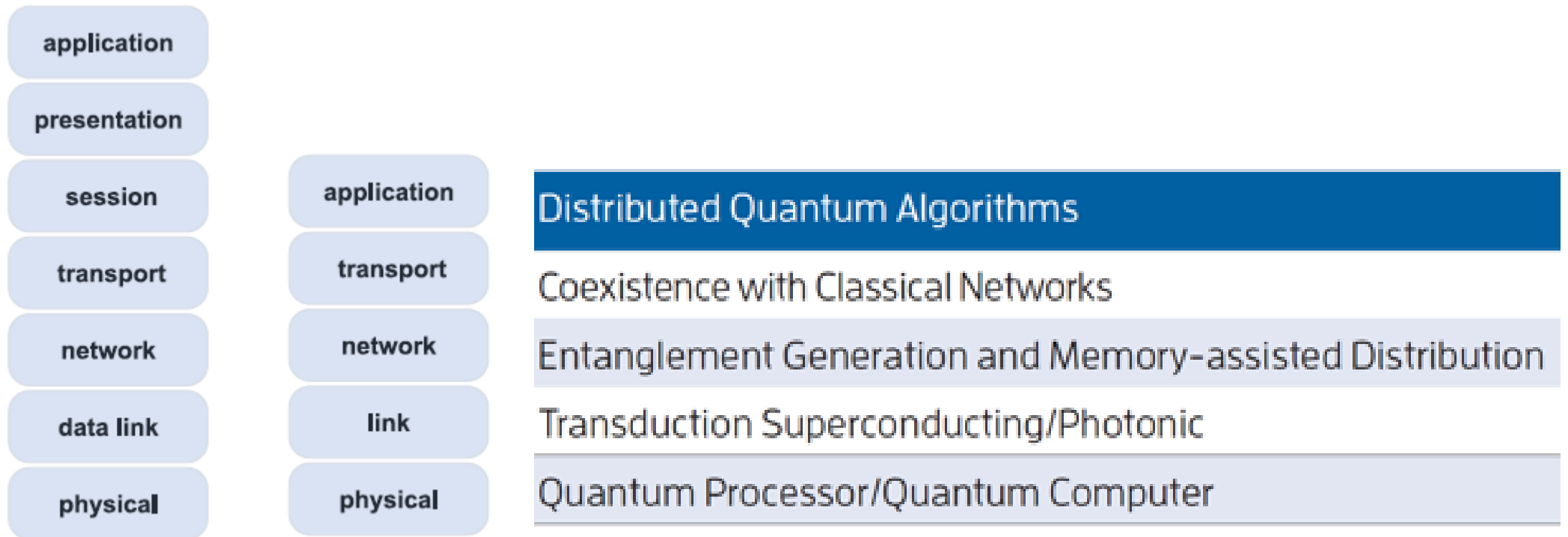
# Quantum internet challenges: everything!

We need to maintain coherence over long-distance (fiber and free-space), at different operating frequencies and synchronized among many users.

- Quantum Internet hardware requirements:
  - Qprocessor & Qmemory with optical interface
  - High-speed, low-loss switches, **repeaters**
  - Improved sources and detectors



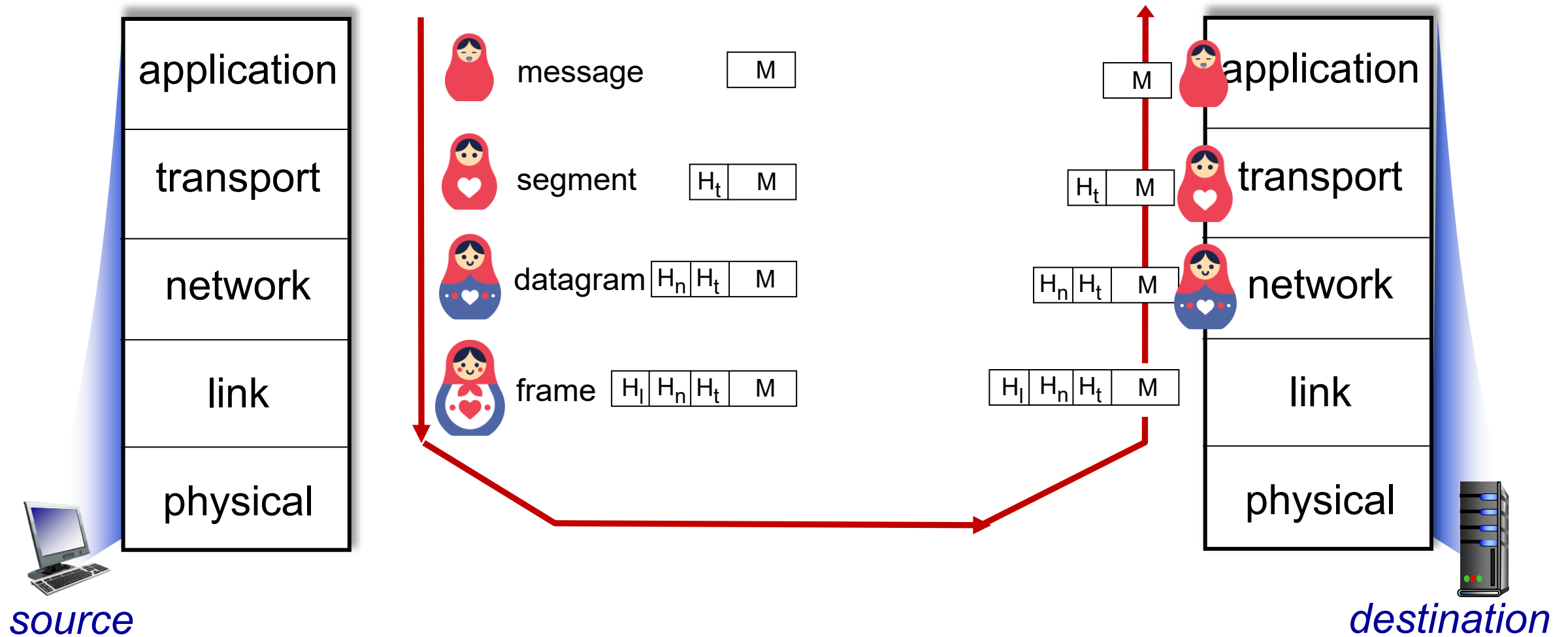
# *Classical vs quantum internet protocol stack*



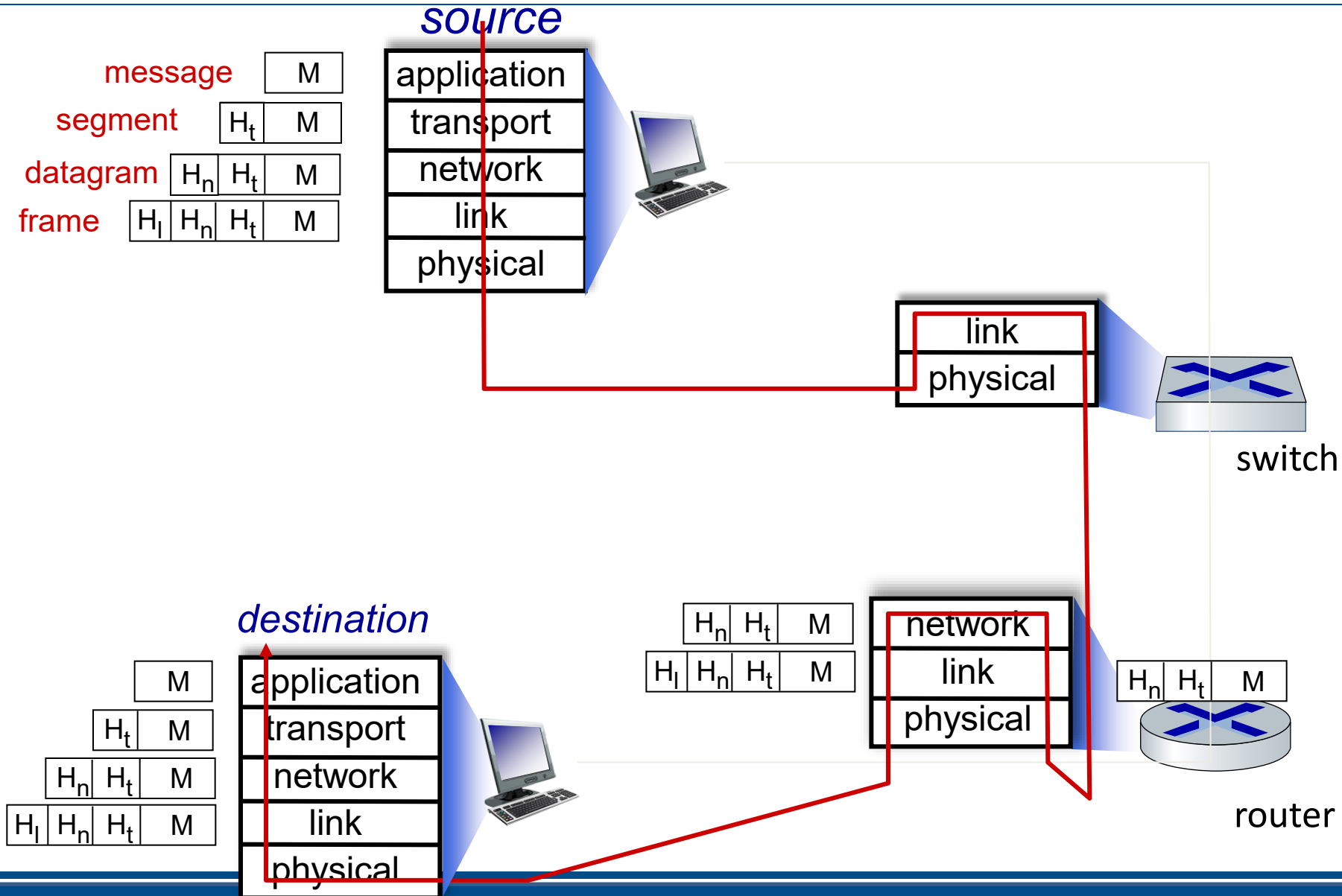
Yuan Li et. al, A Survey of Quantum Internet Protocols From a Layered Perspective, IEEE Comm. Survey & Tutorial, 2024



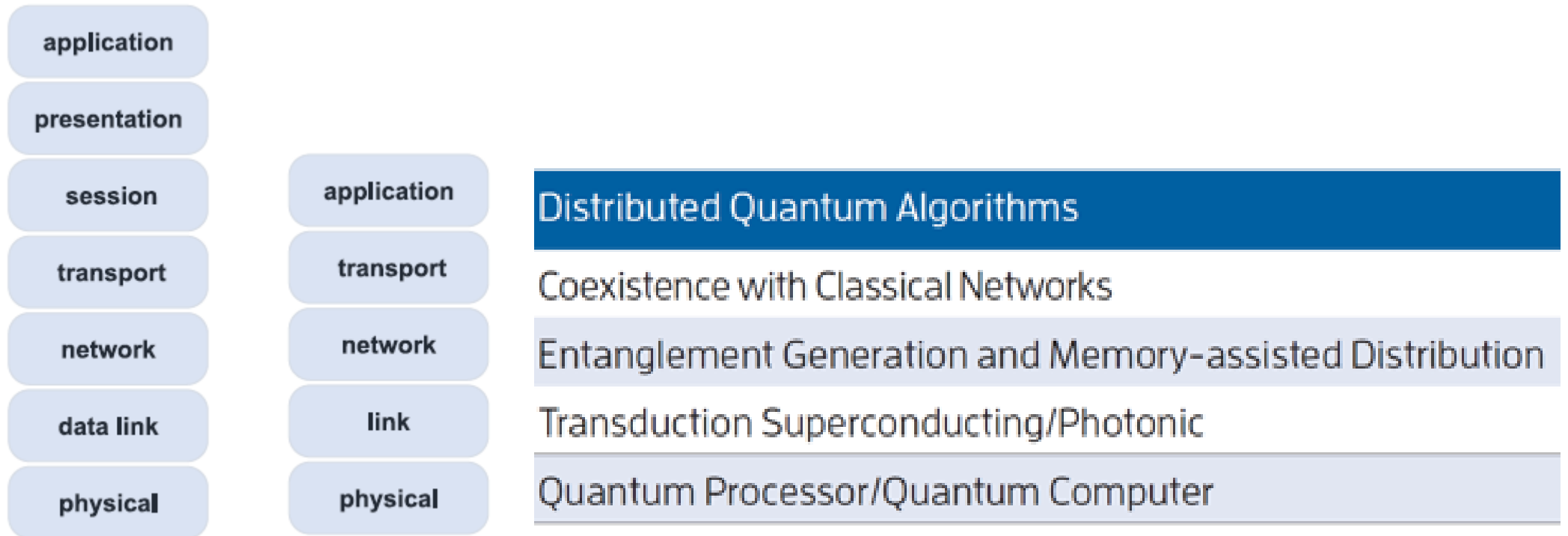
# Classical Internet Protocol Stack



# Classical Networking



# *Classical vs quantum internet protocol stack*



Yuan Li et. al, A Survey of Quantum Internet Protocols From a Layered Perspective, IEEE Comm. Survey & Tutorial, 2024

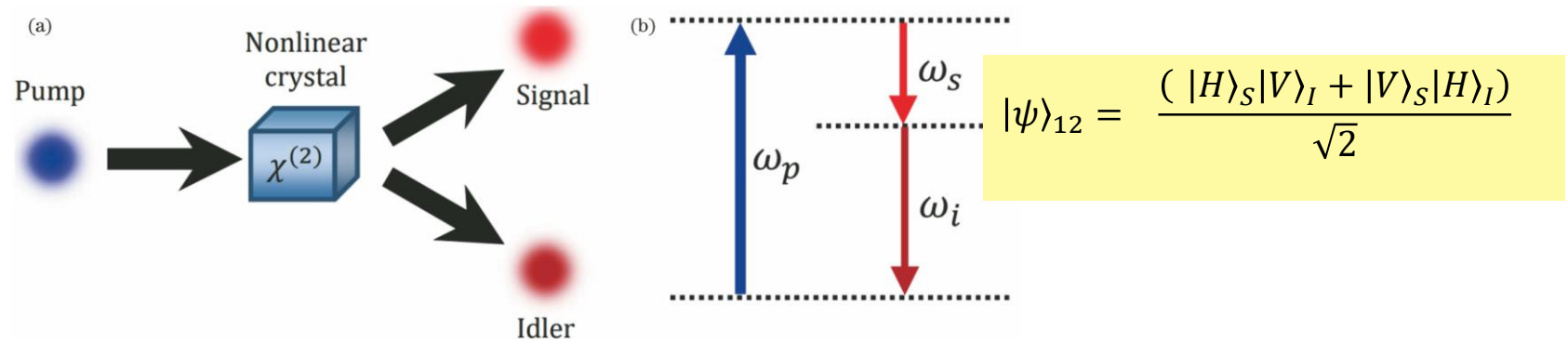
# *Quantum physical and link layers*

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- Quantum physical layer
  - Quantum devices to generate/receive qubits
  - Quantum memory to store qubits
  - Quantum fibers/free-space as propagation medium
- Quantum link layer
  - Generate robust entanglement
  - Quantum transducers for transduction of bits to qubits
  - Coupling of qubits to quantum links
  - Uses error correction codes (QEC)

# How to generate entanglement?

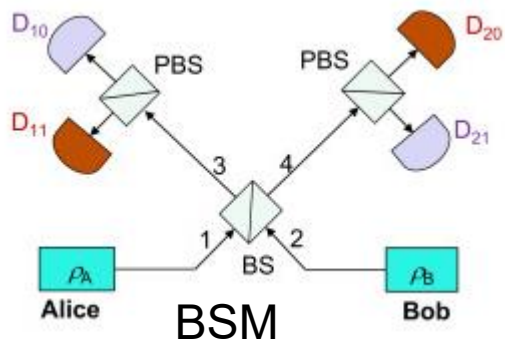
- Single source:



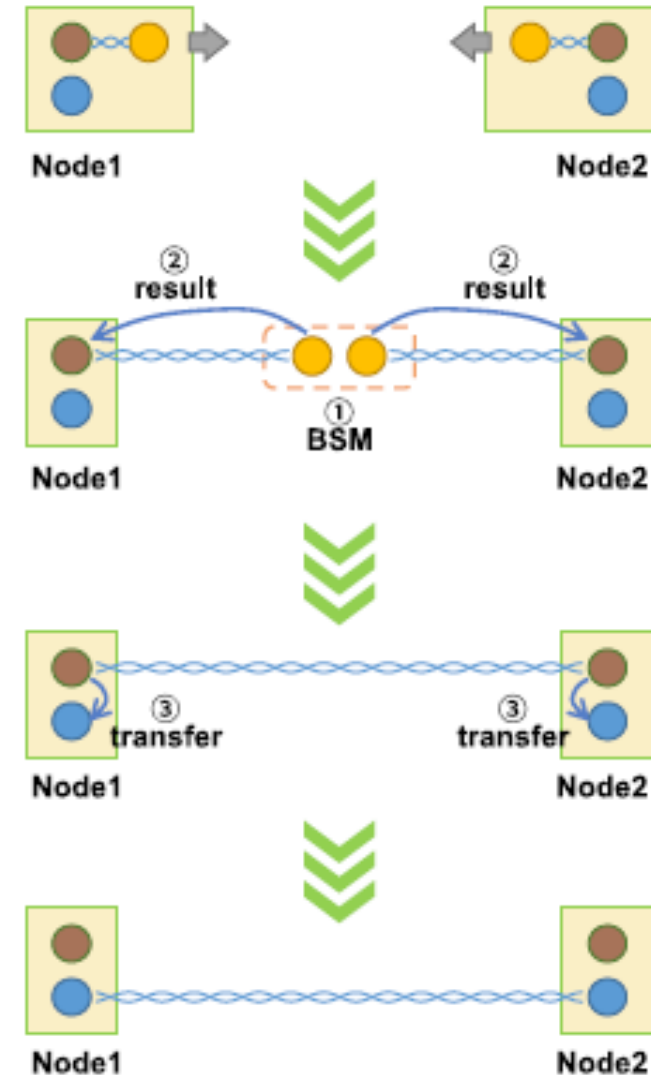
- Between two remote particles (or qubits)?
- Two types of qubits exist within a quantum node:
  - Communication qubits, used for generating entangled states, and
  - Memory qubits, used for storing quantum states.

# MeetInTheMiddle scheme

- First, each node prepares an entangled state between its own communication qubit and a photon.
- Second, the two nodes emit their photons simultaneously to meet each other and perform a BSM at the midpoint between them.
- For the BSM, linear optics and single-photon detectors are employed, for which the success probability is 50% even in the ideal case.
- If the BSM is successful, this indicates that the communication qubits at the two nodes have already formed an entangled state.
- Finally, the state can be transferred from the communication qubits to the memory qubits for better storage



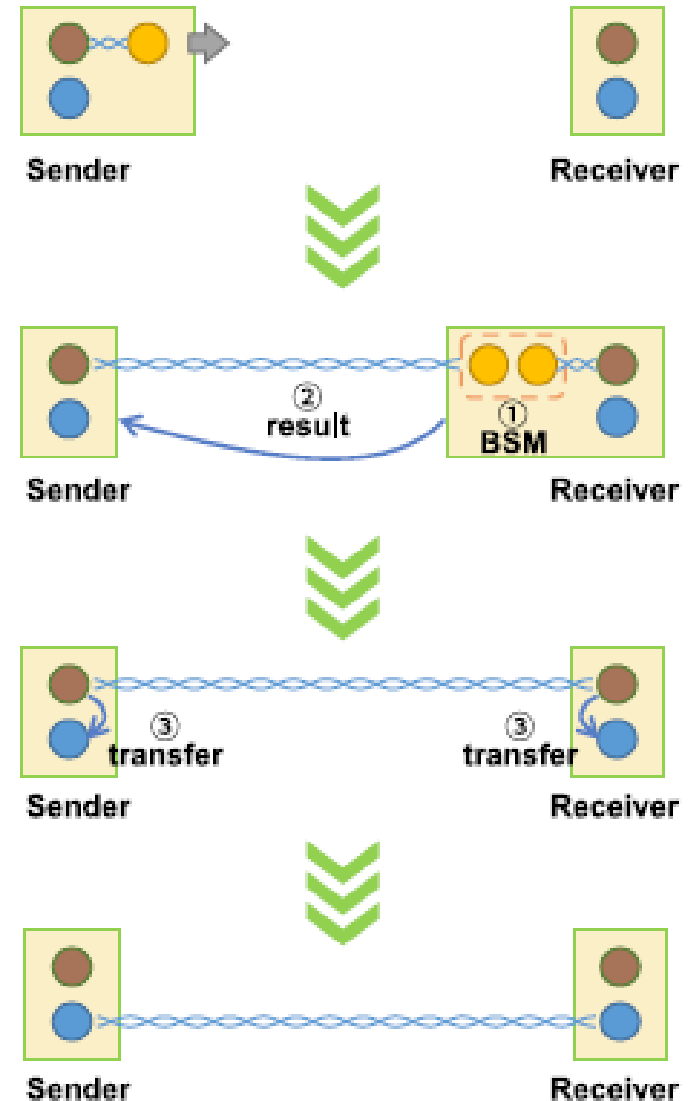
**Brown circles:** communication qubits  
**Yellow circles:** photons  
**Blue circles:** memory qubits  
**Gray arrows:** photon emission.





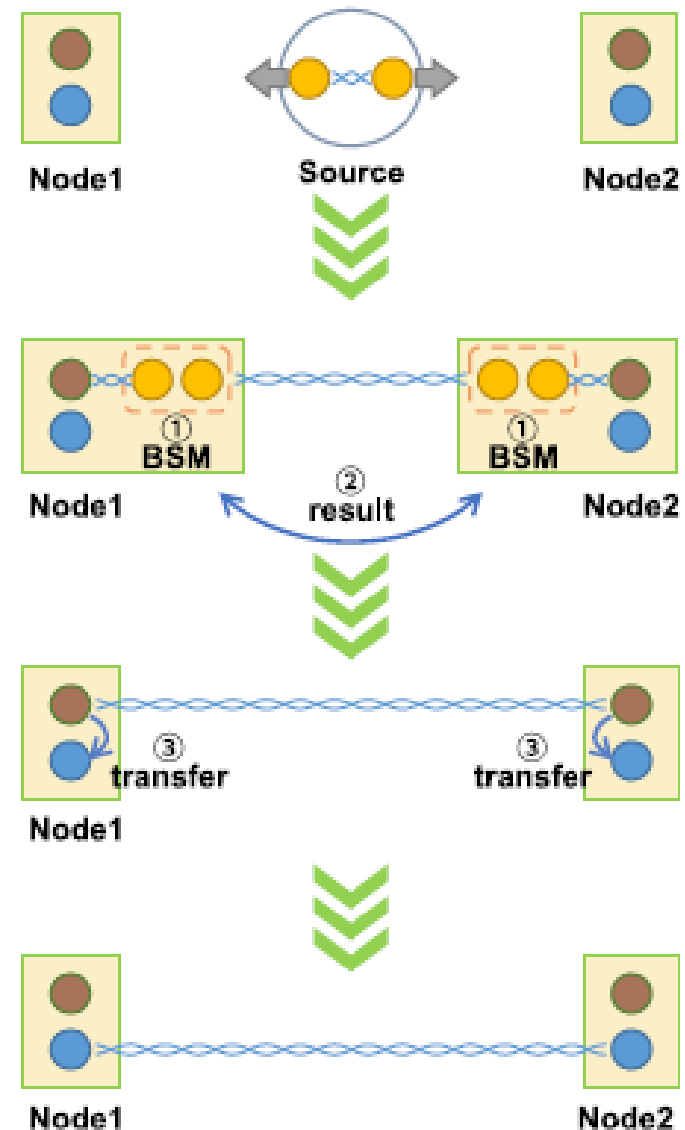
# SenderReceiver scheme

- The sender sends a photon, which is entangled with its communication qubit, to the receiver.
- The receiver prepares an entangled state between its communication qubit and a photon.
- When the photon arrives from the sender, a BSM is executed at the receiver node.
- Then, a classical result message is sent to the sender node for later action.
- In this scheme, the receiver node is immediately aware of success or failure, helping it utilize its qubits effectively.



# MidpointSource scheme

- There is an external light source at the midpoint between the two nodes.
- The source initially sends half of an entangled state to each node.
- Then, the nodes prepare entangled states between their communication qubits and photons.
- When the photons sent from the midpoint arrive, the nodes each need to perform a BSM.
- Then, the nodes send their measurement results to each other for subsequent operations.

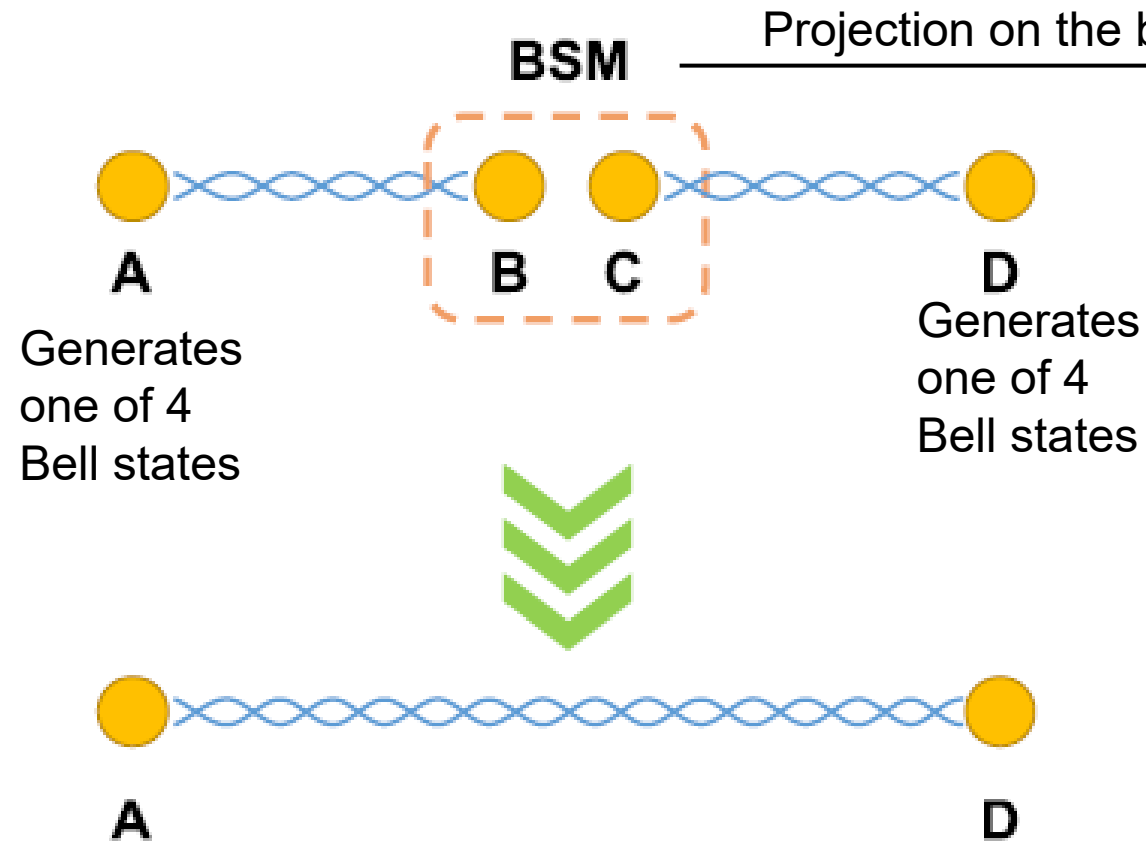


# ***Network layer: entanglement distribution***

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- **How can Alice transmit the state of a qubit  $\Psi$  to Bob far away?**
  - Entanglement swapping
  - Quantum teleportation

# Entanglement swapping



## THE BELL BASIS

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

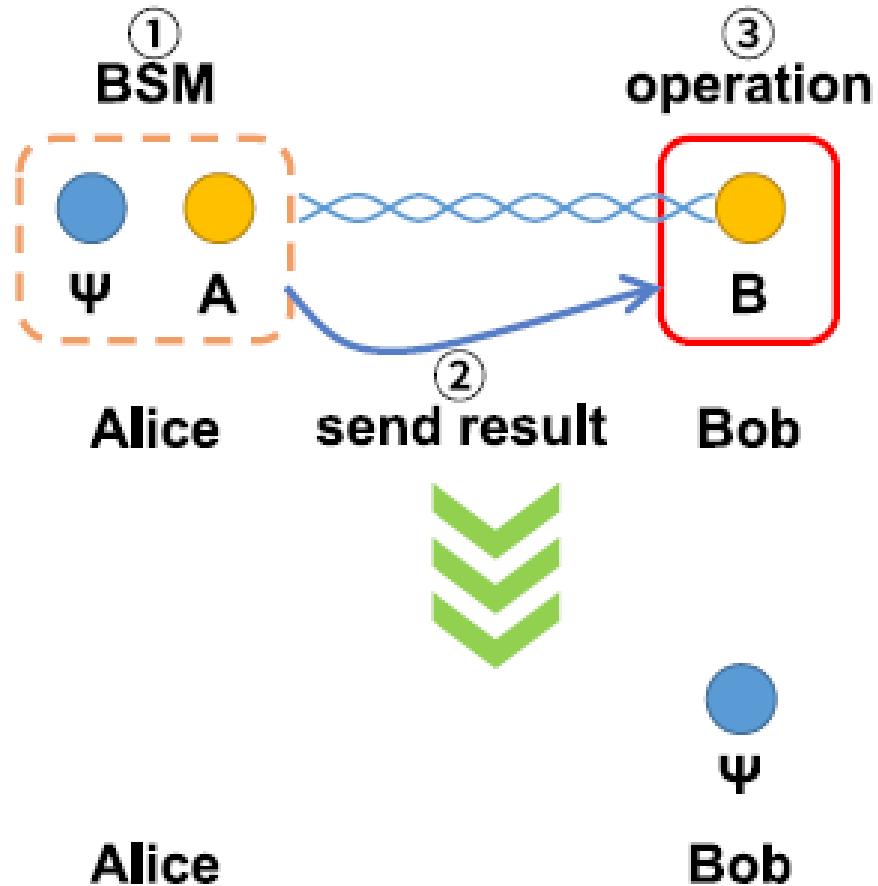
$$|\Phi_-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$|\Psi_+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

$$|\Psi_-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

<https://www.youtube.com/watch?v=1zVvoADzXs4>

# Quantum teleportation



- Filled circles: qubits
- Qubits connected by threaded lines: entangled states
- Solid red rectangle: phase-flip and bit-flip operations
- BSM: Bell-state measurement

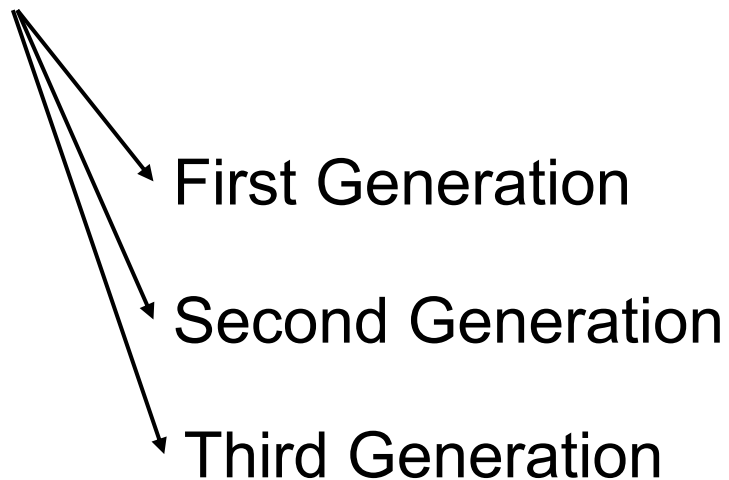
C. H. Bennett et al., “Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels,” Phys. Rev. Lett., vol. 70, no. 13, pp. 1895–1899, Mar. 1993.

# Quantum repeaters

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- Repeater: Regeneration, retiming, and reamplification
- Q Repeater: does not exist in the market, but proposed

Q-repeaters (capability of error correction)



Photon loss &  
Operational errors:  
(due to measurement and  
gates operation)

# *Quantum transport layer protocols*

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- Congestion Control?
- Retransmission?



# Application Layer

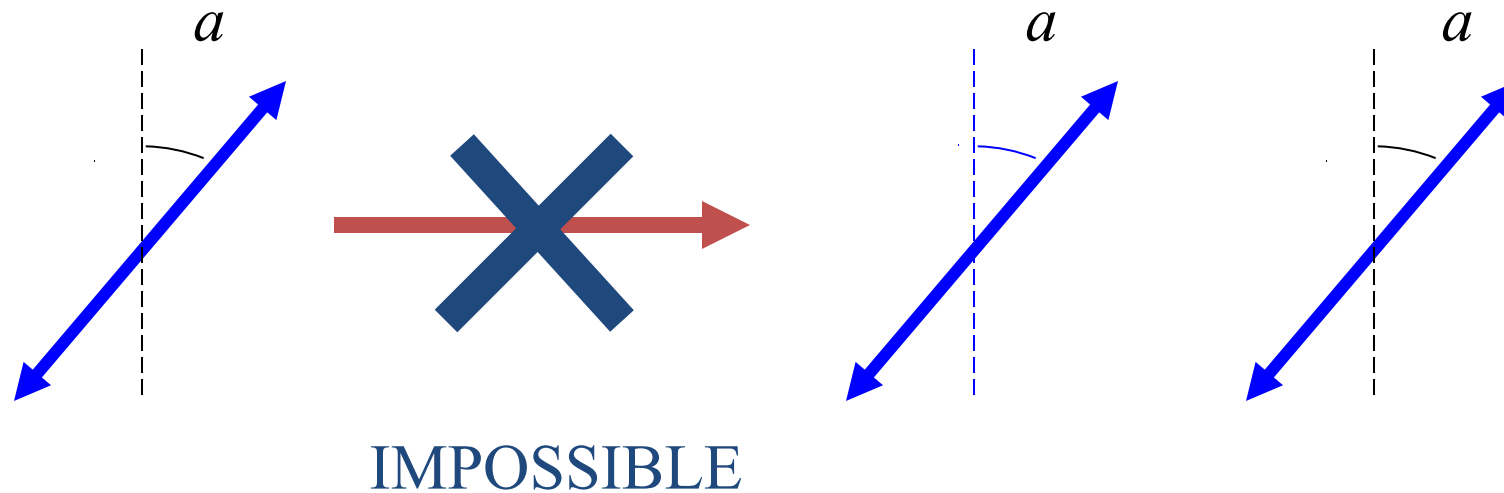
- Example: **quantum key distribution**
- In quantum mechanics, measurements cause disturbance\*
- Can use this disturbance to detect eavesdroppers on a quantum channel
- Leads to provably secure key distribution



Cartoon by John Richardson in Physics World, March 1998

# *Reminder: Quantum No-cloning Theorem*

- An unknown quantum state **CANNOT** be cloned. Therefore, eavesdropper, Eve, cannot have the same information as Bob.
- Single-photon signals are secure.



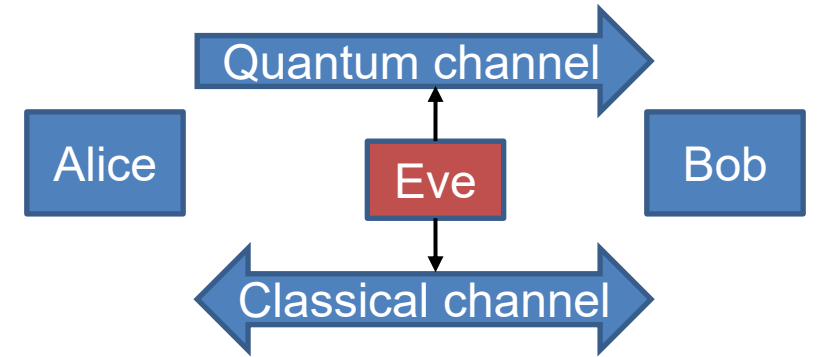
# Bennett and Brassard's scheme (BB84)

- Quantum Part (random key gen)

- Random non-orthogonal signals

- Random measurements

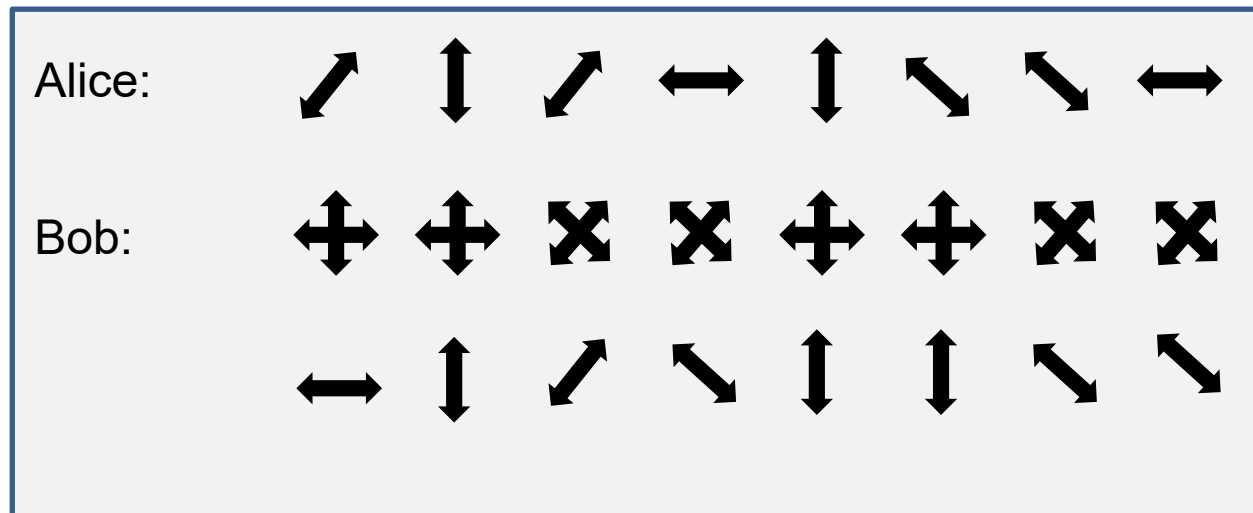
| Basis                                                                               | 0                                                                                   | 1                                                                                   |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |
|  |  |  |



Assumptions:

A, B have access to both channels  
E has complete access to quantum,  
and listen to classical

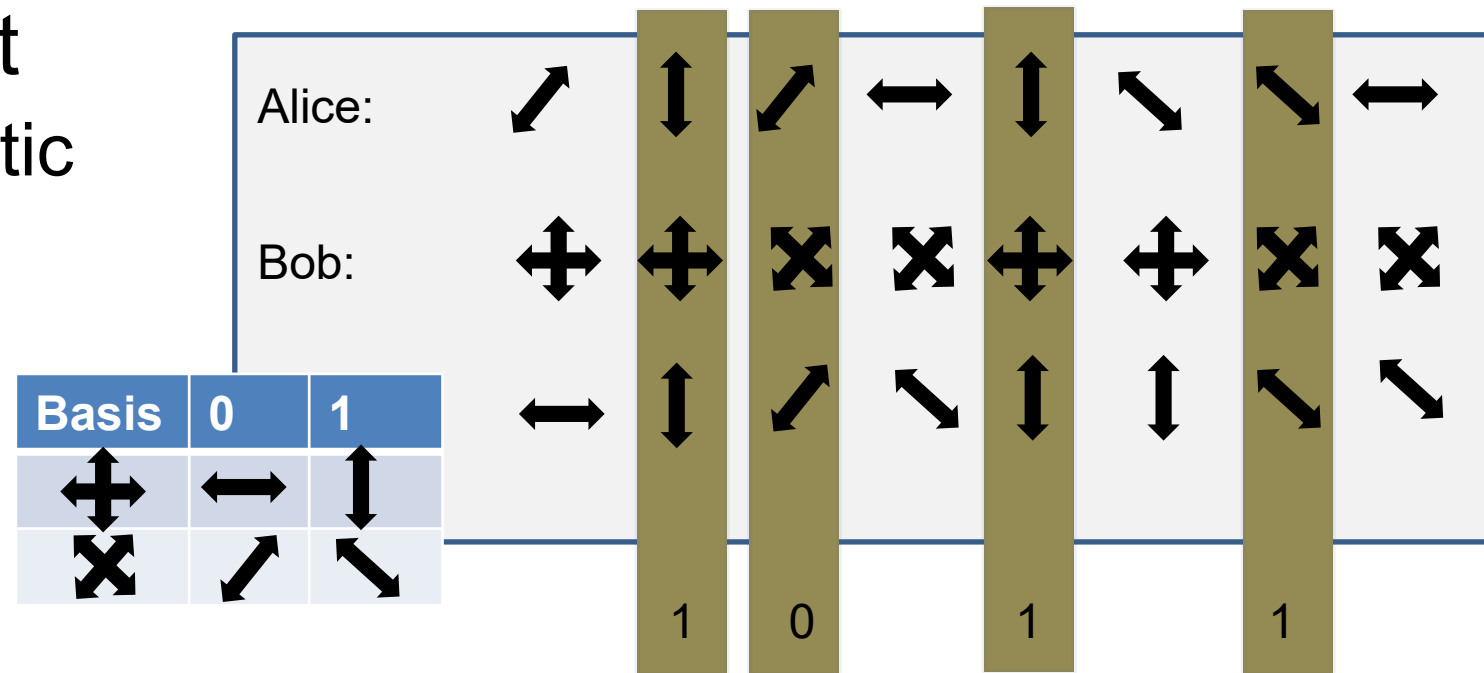
Ideal channel: all errors due to Eve  
Practical: error correction needs to  
be applied



C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," Proceedings of IEEE International Conference on Computers, Systems, and Signal Processing, Bangalore, India, 1984.

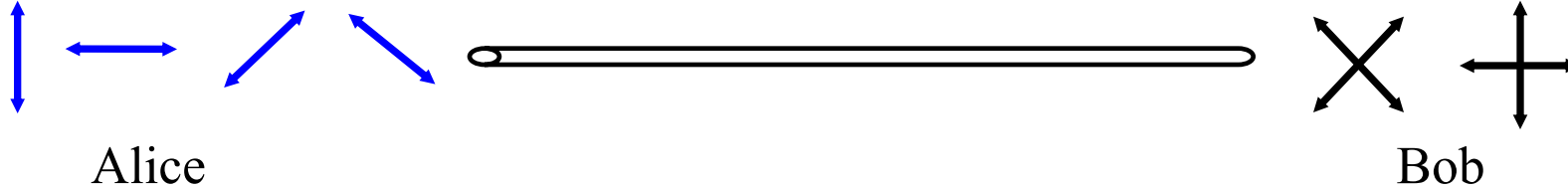
## BB84 contd.

- Classical channel part
  - Distinguish deterministic from random processes
  - Share prepared and measurement bases



- (knowledge of bases does not give Eve clue about trx. signals)
- Statistical test by sharing some data, if error, means eavesdropper in ideal case. Reality: apply error correction

# BB84 Assumption



## ASSUMPTIONS:

1. Source: Emits perfect single photons. (No multi-photons)
2. Channel: noisy but lossless. (No absorption in channel)
3. Detectors: a) Perfect detection efficiency. (100 %)
4. Basis Alignment: Perfect. (Angle between X and Z basis is exactly 45 degrees.)

Assumptions lead to security proofs:

Mayers (BB84), Lo and Chau (quantum-computing protocol),  
Biham et al. (BB84), Ben-Or (BB84), Shor-Preskill (BB84), ...

|                                      |
|--------------------------------------|
| Conclusion: QKD is secure in theory. |
|--------------------------------------|

# *Photon-number splitting attack against multi-photons*

- A multi-photon signal CAN be split. (Therefore, insecure.)

